

but there are many others. As a matter of fact, an ideal NLU system would put millions of people out of work. So why is NLP so big, why is it difficult, what's being done out there and what an ideal NLU can do for us, is what this story is about.

Eliza revisited

Eliza is a primitive NLU program written in the 1960's that emulates the way a psychiatrist speaks. Eliza recognizes simple sentence structures such as subjects and predicates, basic words such as I, go, tell, etc., and a few special words such as family and mother. If you were to type in 'I love my mother', Eliza may reply with, "Who else in your family do you love?" Sounds pretty intelligent, doesn't it? But when you say something it doesn't understand, Eliza just says, "Please go on", or something similar.

Eliza is a good attempt at NLP, but you'll realize its primitiveness when you type in something like, 'My father wins the chad mouse with Greek demons and no chicken soup for Fido'. In this case, Eliza would reply with something like, "Tell me more about your father", because of the word 'father' in that sentence. The problem is, Eliza doesn't understand most of what you're saying.

An ideal NLP, as far as possible, would store information and perform actions that you would expect of a machine that could understand what you are saying given your peculiar speech pattern. Some progress has been made along the direction of getting a machine to understand human language, but we are still very far from where we would like to be.

Why NLP is big

Why is so much serious work being done out there on NLP? What can ideal NLU systems do for us? Think of the possibilities that these systems would open up. They could read newspaper headlines out

IBM's MASTOR

IBM is developing a tool called MASTOR (Multilingual Automatic Speech-to-Speech Translator), which will enable conversation between two people who do not speak a common language. The first speaker speaks into a PDA or laptop, and the other person can hear the translation, spoken out by the PDA, in his or her language.

loud for you, and then give you a little printout of exactly what you are interested in. You'd have to hardly read anything!

And if you are one of those who wade through thousands of lines of technical journals and papers just to gather a few titbits of information, NLU systems could improve your life manifold. They could provide you with a weekly digest, and store information in a continually expanding database. With all the time saved, you would be able to take weekly vacations to Goa!

Need to read a text in Italian? Translate it using Google's patented technology, of course. And how does Google do it? It's not as simple as translating word for word; the system must, to an extent, understand the source text. Unless it understands the context, it translates poorly. For example 'mein Arm' (German for 'my arm'), is translated by Google into 'my lever', as 'Arm' in German can mean both arm and lever.

And wouldn't it be wonderful if you could ask a database to "find me all of Jon's records for the past year, except for the ones in March. Also, no personal information; just the official stuff". Now that would require your machine to understand what you're saying, and is definitely more complex than speech recognition.

Speech recognition has a range of applications beyond being able to tell your computer what to do. Some of them include automated customer service over the phone, training systems—where a computer is the instructor, in say, a distance learning program. In speech, recognition is one aspect, but generation is quite another.

Systems that read out text for visually impaired persons, such as JAWS, can be augmented so that the voice is infused with some emotion, and doesn't sound machine-like. This cannot be done unless



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